DynNavPros: an algorithm for generating navigation prosody motion styles in dynamic environments

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***Abstract*— Although mobile robots may perform useful tasks by navigating in human environments, the relationship between robot motion features and human social perceptions is still not fully understood. Features such as human-robot distance, average velocity or path smoothness have been shown to impact perceptions of comfort, naturalness or human-likeness. More recently, new features called navigation prosody features were proposed, introducing fixed acceleration values and timing-sensitive velocity profile patterns which were found to impact perceptions of *social attitudes* such as confident or hesitant. These features were hard-coded and evaluated in video lab-based experiments due to the absence of real-time navigation algorithms capable of reproducing them. In this paper, we propose the first formalization of the navigation prosody features into a navigation algorithm enabling accurate reproduction of the features in a real-time receding horizon framework. Subsequently, we present the first evaluation of these features in ecological conditions suggesting that the robot's social role during the interaction may alter how the navigation prosody features are perceived, and that additional motion features may also impact perceptions, thereby improving understanding of human social perception of robot navigation.**

I. Introduction

Both in research and industry, many applications involving mobile robots navigating alongside humans are being explored. Advancements in human motion prediction and social navigation algorithms have improved navigation performance in human environments, however the factors which are involved in people's interpretation of the robot's motion are still not fully understood [1]. During human-robot interaction (HRI), robot actions may be perceived as conveying *social attitudes* such as polite or authoritative, aggressive or gentle, confident or hesitant [2], [3], [4]. This kind of social perception can form part of people's evaluation of robot systems, and even influence the person's own actions [5], [6].

Typical studies in social navigation concern themselves with evaluating if the overall motion generated by the algorithm is perceived as comfortable, natural, legible or safe [7]. In contrast, we aim to understand the contribution of individual motion features to human perception of robots, specifically with regard to perceptions of *social attitudes*. In a recent work [3], previously untested motion features consisting of subtle temporal patterns in velocity profile acceleration values were found to significantly affect human perception of social attitudes when evaluated in video and lab-based perception tests using hard-coded motions. Current social navigation and expressive motion generation algorithms do not provide control over these motion features [1], [7]. As a result, these motion features have never been studied in ecological conditions, making it unclear to what extent they may contribute to perceptions of robots in real-world deployment. In order to understand the impact of an individual motion feature on such perceptions, it is also important that the algorithm enables different subsets of features to be selected, and that the resulting motion corresponds to a known combination of the motion features enabling accurate comparisons of motion styles.

In this paper, we propose the first algorithm to provide control over the set of so-called navigation prosody features proposed in [3] while navigating in dynamic environments, and present the results of the first experiment testing these feature's impact on HRI in an ecological experiment. The contributions of this paper are:

- novel timing-aware constraints taking the navigation prosody features proposed in [3] from hard-coded velocity profiles to real-time navigation;
- modular constraint hierarchy balancing motion feature preservation with safety in uncertain environments;
- first analysis of the effects of prosody features on human perceptions of a mobile robot deployed with a real task and social role.

II. Related work

A. Expressive robot motion

Expressive robot motion has often been studied using hand-designed motions to convey emotions [8], [9], [10]. Instead, in [3], subtle differences in timing and acceleration and velocity based features of a mobile robot lead to changes in perceptions of *social attitudes* (e.g. confident/hesitant, aggressive/gentle) and physical characteristics of the robot (e.g. sturdy/frail). A smaller subset of works explore algorithms for functional expressive motions [11], [12], [13], [14], aiming to accomplish a practical task while maintaining control over the expression of the robot's goal or emotions. Such approaches are usually evaluated in static environments, and the impact of real-time re-planning required to handle uncertainty in dynamic environments is not considered. Learning-based methods for emotion expression have also been proposed [15] where the motion features are implicitly represented by a neural network, lacking a means to selectively include individual motion features. Our goal is to propose an algorithm enabling different combinations of the novel motion features proposed in [3] to be reproduced under real-time re-planning constraints to evaluate their impact on social perceptions of mobile robots in realistic conditions.

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B. Social Navigation

Social navigation (SN) works are mostly framed with the goal of enabling robots to accomplish their practical tasks of navigation in complex dynamic environments such as crowd navigation [16]. In [17], factors related to the complexity of social navigation scenarios are proposed pertaining to the physical complexity of crowd navigation, calling for human *social perception* of the robot's behaviours to be studied as future work. When evaluating SN algorithms, human perception is considered in terms of perceived comfort, naturalness, and high-level norms such as on which side of a corridor a robot should navigate [7]. Algorithms based on machine learning methods [18], [19], or trajectory optimization with additional cost functions [20] produce a single style of robot navigation which is designed to be unobtrusive and efficient. In [21], a VLM is used to determine contextually appropriate navigation actions, evaluated according to success rate, collisions, and perceived comfort, safety and legibility. In contrast, we aim to design an algorithm that can be configured to produce an array of different motion styles enabling the effects of specific, formalized individual motion features to be compared. Furthermore, we aim to control the navigation prosody features presented in [3] which differ from existing features, in order to assess their effect on HRI during real-world dynamic navigation scenarios.

C. Background: navigation prosody features

In this section we present the set of motion features proposed in [3]. Piecewise linear velocity profiles are shaped according to three categories of features.

1) Kinematics types: These features constrain the acceleration a_{kin} and maximum velocity v_{kin} to an exact value. There are three different settings designated as *low*, *medium* or *high* kinematics types, using values $a_{kin} = 0.2$, 0.35 or $0.5m.s^{-2}$ and $v_{kin} = 0.25$, 0.5, or $0.75m.s^{-1}$, respectively.

2) Motion Sequences: These features involve introducing specific acceleration patterns under certain conditions. The *pause* motion sequence requires that the robot maintain a constant velocity for $t_{pause} = 300ms$ when transitioning between acceleration and deceleration (Fig. 1a). The *hesitation* motion sequence requires repeating a $t_h = 600ms$ deceleration and acceleration (Fig. 1b) at regular time intervals $t_{h_int} = 2000ms$.

3) Variants: These features involve altering the local quality of the velocity profile. An acceleration under the *smooth* variant consists of a single affine segment with acceleration a_{kin}. Under the *saccade* variant, small amplitude high frequency oscillations are introduced to achieve a stuttering, jittery motion. These are modeled with a triangle wave with a period $\pi = 0.02s$, and an amplitude A dependent on the kinematics type: $A_{low} = 0.02m.s^{-2}$, $A_{medium} = 0.05m.s^{-2}$, $A_{high} = 0.07m.s^{-2}$. Under the *increment* variant, two constant velocity segments are introduced at $v = \frac{1}{3}a_{kin}$ and $v = \frac{2}{3}a_{kin}$, creating a stepped acceleration (Fig. 1c).

These motion features were derived by analogy to vocal prosody whereby small changes in voice can alter social perceptions between humans [22], [23] and with robots [24], hence they are referred to as features of *navigation prosody* [3]. These features were hard-coded into velocity profiles for a video-based experiment and a lab-condition experiment where a mobile robot approached a person without any specific task or role, resulting in mappings between the feature combinations and ten social attitude scales such as the confident-hesitant scale (Table I). In this work, we aim to formalize the characteristics of each motion feature and integrate them into a real-time receding horizon control algorithm to enable the effects of different combinations of these motion features to be tested in real ecological scenarios, beyond hard-coded lab experiments.

TABLE I
EXAMPLE RELATIONS BETWEEN MOTION FEATURES AND SOCIAL ATTITUDE PERCEPTIONS BASED ON [3].

Social attitude	**Motion features**
Confident	Motion sequence: pause Kinematics type: high Variant: smooth
Hesitant	Motion sequence: hesitation Kinematics type: low Variant: saccade

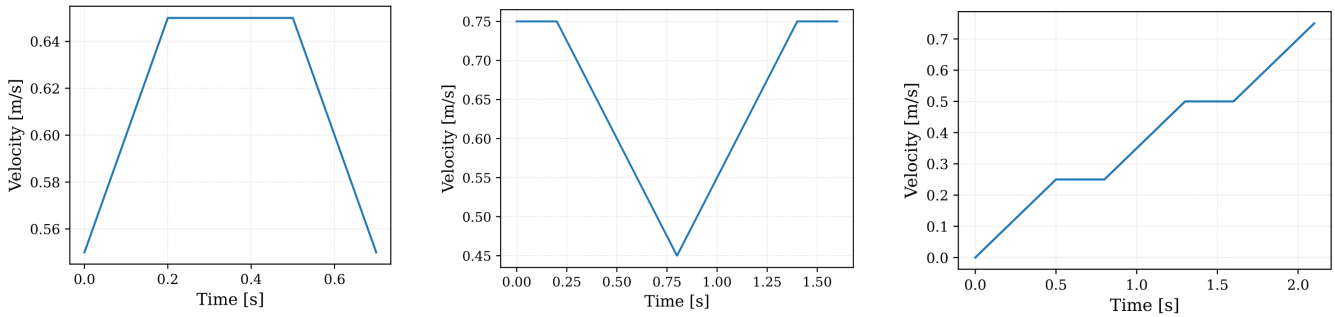
III. NAVIGATION ALGORITHM FOR CONTROLLABLE PROSODY

In this section, we present our approach to design a navigation algorithm capable of re-planning trajectories in dynamic uncertain environments, while ensuring they accurately reproduce the motion features corresponding to social attitude expressions presented in the previous section.

The piecewise-linear velocity profiles and motion features can be represented as a sequence $U = (u_0, u_1 \dots u_{N-1})$ of N motion phases consisting of an acceleration $a_k \in \mathbb{R}$, and duration $t_k \in \mathbb{R}_{>0}$, i.e. $u_k = (a_k, t_k)$. In conjunction with an initial position $x_0 \in \mathbb{R}$ along the robot's forward axis, and initial linear velocity $v_0 \in \mathbb{R}$, these values determine the robot's trajectory in space and time, and are related through the forward kinematics relation (1).

$$\begin{aligned} x_{k+1} &= x_k + v_k t_k + \frac{1}{2} a_k t_k^2 \\ v_{k+1} &= v_k + a_k t_k \end{aligned} \tag{1}$$

We formulate an optimization problem (2) where the lengths and slopes of the motion phases are optimized to minimize distance to the goal position, subject to a set of novel prosody constraints (Section III-A) modeling the navigation prosody features presented in Section II-C. We chose to enforce the features through constraints to keep a clear separation between the motion style and the navigation objective function, enabling deployment in different navigation experiment scenarios without systematic re-tuning of the objective function. Changing the subset of active constraints PC_{active} enables the range of motion feature combinations studied in [3] to be reproduced. Optimization is performed in



(a) Pause motion sequence. (b) Hesitation motion sequence. (c) Increment variant.

Fig. 1. Illustration of several navigation prosody motion features.

a receding horizon fashion over a fixed horizon T_h, subject to additional collision avoidance and terminal constraints (Section III-B).

$$\min_{u_0 \dots u_{N-1}} \quad \sum_{k=0}^{N-1} ||x_g||^2$$
$$\text{s.t.} \begin{cases} \text{all constraints } C \in PC_{\text{active}} \text{ satisfied}, \\ \sum_{k=0}^{N-1} t_k = T_h \,, \\ CollCritical \quad (19)\,, \\ CollIdeal \quad (20)\,, \\ TerminalCons \quad (21)\,. \end{cases} \tag{2}$$

A. Prosody constraints

We design constraints that enforce consistent reproduction of the motion features presented in Section II-C. Each constraint is formalized as a logical predicate evaluating to $true$ when the trajectory matches the motion feature.

1) Kinematics types: This feature is formalized through constraints on the admissible values for acceleration (3) and velocity (4). Note that accelerations are restricted to a discrete set of three values, as opposed to being continuous and bounded. The **high**, **medium**, and **low** kinematics types are implemented by assigning different values to a_{kin} and v_{kin} (Section II-C).

$$a_k \in \{-a_{kin}, 0, a_{kin}\} \tag{3}$$

$$0 \leq v_{k+1} \leq v_{kin} \wedge \neg(a_k = 0 \wedge v_{k+1} \notin \{v_{kin}, 0\}) \tag{4}$$

$$KinematicsCons(U) \iff (3) \wedge (4) \tag{5}$$

Several motion features require successions of motion phases to be performed with a given acceleration value and duration. When re-planning trajectories dynamically, the constraints must enforce these rules which may depend on the history of motions performed by the robot. We introduce a new variable representing the cumulative duration of a motion phase denoted as $t_cumul_k \in \mathbb{R}_{>0}$ (6).

$$t_cumul_k = \sum_{i=j}^{k} t_i, \quad s.t. \quad \forall i \in [j,k]\,, a_i = a_k \tag{6}$$

2) Motion sequences: The **pause** motion sequence consists in ensuring acceleration and deceleration phases are always separated by a constant velocity phase with $t_k \geq t_{pause} = 300ms$. We formalize the constraint in (9) as the combination of two predicates modeling invalid trajectory features. Equation (7) models a change in acceleration not being followed by the required constant velocity phase. Equation (8) enforces that a previously initiated pause phase cannot be shortened or left incomplete, ensuring conservation of the pause motion sequence under dynamic re-planning.

$$\begin{aligned} &a_{k-1} \neq 0 \ \wedge \ a_k \neq a_{k-1} \\ &\wedge \ \neg(a_k = 0 \ \wedge \ t_k \geq t_{pause}) \end{aligned} \tag{7}$$

$$\begin{aligned} &(a_{k-1} = 0 \ \wedge \ t_{k-1} < t_{pause}) \\ &\wedge \ \neg(a_k = 0 \ \wedge \ t_cumul_k = t_{pause}) \end{aligned} \tag{8}$$

$$PauseCons(U) \iff \neg((7) \vee (8)) \tag{9}$$

The **hesitation** motion sequence consists in introducing a short deceleration and acceleration at regular intervals $t_{h_int} = 2000ms$. We introduce the variable $type_k \in \{\text{hesit}, \text{normal}\}$ to distinguish between phases constituting part of a hesitation sequence (hesit) from normal phases (normal). Equation (10) models that a hesitation deceleration should be planned after a normal acceleration, or once the time elapsed since the last hesitation $t_{since_h} \in \mathbb{R}_{>0}$ reaches t_{h_int}. Equation (11) models that a hesitation deceleration should be followed by a hesitation acceleration. Equations (12) and (13) account for re-planning by using the robot's motion history to ensure partially executed hesitation phases are completed by the next motion phase.

$$\begin{aligned} &\Big[\big(a_{k-1} = a_{kin} \wedge \ type_{k-1} \neq \text{hesit}\big) \ \vee \ t_{since_h} \geq t_{h_int}\Big] \\ &\wedge \Big[(a_k = -a_{kin} \ \wedge \ t_k \neq t_h) \ \vee \ a_k = 0\Big] \end{aligned} \tag{10}$$

$$\big(a_{k-1} = -a_{kin} \ \wedge \ type_{k-1} = \text{hesit} \ \wedge \ t_cumul_{k-1} = t_h\big) \\ \wedge \neg\,(a_k = a_{kin} \ \wedge \ t_k = t_h). \tag{11}$$

$$\big(a_{k-1} = -a_{kin} \ \wedge \ type_{k-1} = \text{hesit} \ \wedge \ t_cumul_{k-1} < t_h\big) \\ \wedge \ \neg\,(a_k = -a_{kin} \ \wedge \ t_cumul_k = t_h). \tag{12}$$

$$\big(a_{k-1} = a_{kin} \ \wedge \ type_{k-1} = \text{hesit} \ \wedge \ t_cumul_{k-1} < t_h\big) \\ \wedge \ \neg\,(a_k = a_{kin} \ \wedge \ t_cumul_k = t_h). \tag{13}$$

$$HesitationCons(U) \iff \neg((10) \vee (11) \vee (12) \vee (13)). \tag{14}$$

3) Variants: The **smooth** variant consists in avoiding jolts or oscillations in the velocity profile, by imposing a minimum duration for each motion phase $t_{smooth} = 300ms$. This constraint formulation also requires considering the cumulative duration of the motion phase, in order to account for the fact that a plan starting with a motion phase shorter than t_{smooth} is still valid if it was preceded by a long motion phase with the same acceleration value (15).

$$SmoothCons(U) \iff t_cumul_k \geq t_{smooth} \tag{15}$$

The **saccade** variant is formalized by initially planning under the smooth constraint (15), and subsequently adding a time-varying velocity offset to the resulting velocity profile. The offset produces oscillations of which the period and amplitude are given in Section II-C.

The **increment** variant requires insertion of constant velocity phases similar to the pause constraint, so we re-use part of the pause constraint to ensure that a partially executed pause is finished by the next motion phase (8). Equation (16) ensures that all motion phases end at a valid velocity, i.e. a multiple of $v_{inc} = \frac{1}{3}v_{kin}$. Equation (17) models the absence of a pause after an acceleration or deceleration phase ending at one of the increment velocities.

$$v_{k+1} = i * v_{inc}, i \in \mathbb{N} \tag{16}$$

$$(a_{k-1} \neq 0 \wedge v_k = i * v_{inc}, i \in \mathbb{N}) \\ \wedge \neg(a_k = 0 \wedge t_cumul_k = t_{pause}) \tag{17}$$

$$IncrementCons(U) \iff (16) \wedge \neg((17) \vee (8)) \tag{18}$$

B. Constraint hierarchy

If the robot is already in the middle of performing certain fixed-duration features like a hesitation or pause when a large unpredicted change occurs in the environment, the optimization problem may become infeasible. Taking an all-or-nothing approach to the motion style enforcement leaves no alternative but to use a fall back method such as emergency braking, which violates all prosody constraints. Instead, we propose to use a hierarchy of constraints given in Table II, enabling various levels of constraint satisfaction. If no trajectories satisfy all constraints, we drop the constraints associated with the lowest priority level, enabling more solutions to be explored in order to maintain partial reproduction of the desired navigation style. We elected to discard entire constraints while fully maintaining the remaining constraints in order to ensure that at any given time, we can express the current style as a combination of the navigation prosody motion features [3], rather than allowing each motion feature to be slightly altered.

TABLE II

CONSTRAINT HIERARCHY

Priority level l	Constraints C
1 (highest)	Collision Critical (19), Terminal (21), Kinematics (5)
2	Smooth (15), Increment (18), Hesitation (14), Pause (9)
3 (lowest)	Collision Ideal (20)

In cases where a planned trajectory is close to a collision state, small inaccuracies in prediction or sensing inherent to real-world deployment may result in unnecessary discarding of prosody constraints. To further increase the likelihood of all prosody features being maintained, we introduce a high priority constraint with robot-obstacle distance $d_{critical} \in \mathbb{R}_{>0}$ which ensures there are no collisions, while a low priority constraint with robot-obstacle distance $d_{ideal} > d_{critical}$ ensures that trajectories will initially be planned with a greater obstacle distance margin. We formulate a simple collision constraint (19) by checking whether the difference between the distance to the closest obstacle d_{obs} and the robot's future position x_k is greater than a threshold $d_{critical}$.

$$CollCritical(U) \iff d_{obs} - x_k > d_{critical} \tag{19}$$

Similarly, the ideal obstacle distance constraint is given in Equation (20). By separating these two constraints into different priority levels of the hierarchy (Table II), if the planner is unable to find a trajectory with $d_{obs} > d_{ideal}$, then it may plan prosody-compliant trajectories where $d_{critical} \leq d_{obs} \leq d_{ideal}$.

$$CollIdeal(U) \iff d_{obs} - x_k > d_{ideal} \tag{20}$$

We adopt a terminal constraint that forces the robot's velocity to be zero at the end of the trajectory (21). When using a terminal constraint, the relationship between the robot's maximum velocity and deceleration dictate a minimum horizon duration T_h required to be able to plan trajectories at maximum velocity. In our approach, we choose a longer T_h, such that the robot plans trajectories enabling it to safely stop *while satisfying the prosody constraints*.

$$TerminalCons(U) \iff v_{N-1} = 0 \tag{21}$$

C. Implementation

In this section we describe the approach we use to solve the optimization problem (2). We approach the optimization problem by discretizing the action space. The accelerations a_k are constrained to one of three discrete values by the kinematics constraint (3). We discretize the durations t_k of each motion phase choosing $dt = 100ms$ to ensure feasibility of all motion features. The planning horizon is set to $T_h = 40dt$, and the number of motion phases is set to $N = 16$. This provides enough time and flexibility to plan the longest motions resulting from the increment and hesitation constraints. Motion phase durations are bounded by t_{max} (22). The collision constraint thresholds are empirically set to $d_{critical} = 0.05m$ and $d_{ideal} = 0.2m$.

$$t_{max} = T_h - \sum_{i=0}^{k} t_i - (N-k)dt \tag{22}$$

TABLE III

NOTATIONS FOR THE TRAJECTORY GENERATION ALGORITHM.

Symbol	Meaning
x_g	Goal position
H	Constraint hierarchy (Tab. II)
PF	Prosody feature subset
$\mathbf{x}_k$	State at phase k: $(x_k, v_k, t_{since_h}, t_{cumul}, d_{obs})$
u_k	k^{th} phase (a_k, t_k)
U^*	Optimal sequence $(u_0, \ldots, u_{N-1})$
$\mathcal{A}$	Acceleration set $\{a_{kin}, 0, -a_{kin}\}$
$\mathcal{T}$	Duration set $\{dt, 2dt, \ldots, t_{\max}\}$
T	Trajectory tree
l_c, l_t, l_{best}	Priority levels for the current phase, current trajectory, and best trajectory

Alg. 1 gives an overview of how the proposed prosody constraints and hierarchy are integrated into a trajectory optimization algorithm (notations in Table III). First, the set of active prosody constraints is constructed based on the desired social attitude to be expressed (lines 1-7). Hesitant and confident attitudes are used to illustrate the process, however all social attitudes from [3] can be modeled by different combinations of our constraints. A tree of trajectories is then constructed, where each node represents a robot state, and each edge represents a motion phase $u_k = (a_k, t_k)$. The tree is expanded by enumerating combinations of accelerations and durations for the N motion phases. Prosody and collision constraints are evaluated on the resulting states in order to determine the motion phase's priority level in our constraint hierarchy (lines 14-15). Motion phases which satisfy fewer constraints than the current best full trajectory are not added to the tree. The priority level of each trajectory is computed once its final motion phase is added. Subsequently, the objective function given in (2) is evaluated on the trajectories satisfying the highest level of the constraint hierarchy, yielding the optimal trajectory. Lastly, the first acceleration command u_0^* is executed, with an additional offset if the saccade variant is active. This process is repeated in a receding horizon fashion at a rate of $10hz$.

Algorithm 1 Trajectory generation with controllable social attitude expression

Input: $\mathbf{x}_0$ initial state; x_g; H ; PF
Output: $u_0 = \{a_0, t_0\}$ applied control command

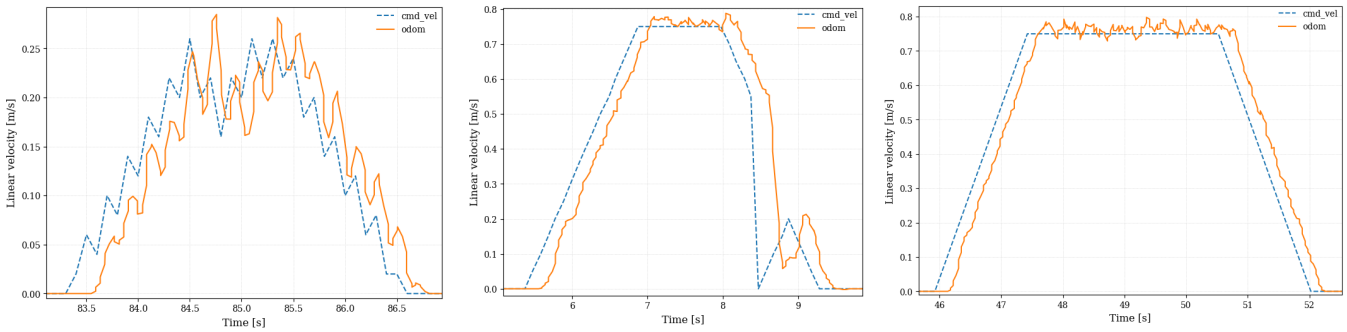
1: **if** $PF =$ Hesitant **then**
2: $PC_{active} \leftarrow \{(5), (14), (15)\}$ ▷ Table I
3: $a_{kin}, v_{kin}, saccades \leftarrow a_{low}, v_{low}, true$
4: **else if** $PF =$ Confident **then**
5: $PC_{active} \leftarrow \{(5), (9), (15)\}$ ▷ Table I
6: $a_{kin}, v_{kin}, saccades \leftarrow a_{high}, v_{high}, false$
7: **end if**
8: $l_{best} = 1$
9: **while** GoalNotReached **do**
10: $T \leftarrow$ InitializeTree($\mathbf{x}_0$)
11: **for** each depth $k = 0 \ldots N-1$ **do**
12: **for** each node n at depth k in T **do**
13: **for** each motion phase $u_k \in \mathcal{A} \times \mathcal{T}$ **do**
14: $\mathbf{x}_{k+1} \leftarrow$ ForwardSimulate(n, u_k)
15: $l_c \leftarrow$ EvalConstr($\mathbf{x}_{k+1}, u_k, PC_{active}, H$)
16: **if** $l_c \geq l_{best}$ **then**
17: AddChild($T, n, \mathbf{x}_{k+1}, l_c, u_k$)
18: **end if**
19: **if** $k == N-1$ **then**
20: $l_t \leftarrow$ TrajectoryPriorityLevel(T, n)
21: $l_{best} \leftarrow max(l_{best}, l_t)$
22: **end if**
23: **end for**
24: **end for**
25: **end for**
26: $U^* \leftarrow$ SelectOptimalTrajectory(T, x_g, l_{best})
27: ExecuteAcceleration(u_0^*, saccades)
28: **end while**

The algorithm was implemented as a ROS node in C++, with one control update cycle taking $80ms$ on average on a single thread of an $i5-8350u$ tablet processor. Person and obstacle detection required for real-world experimentation were performed using a 2D lidar sensor with off-the-shelf methods. Orientation corrections were performed using a proportional controller rate-limited to a_{kin} and v_{kin}. We applied corrections to the robot's state to account for delays in sensing and motor response, enabling accurate planning and execution.

IV. EXPERIMENTS

In this section, we firstly evaluate our algorithm's ability to accurately reproduce the desired motion features. Secondly, we compare the effect of motion features produced by our algorithm in an *in-the-wild* HRI experiment[1] to the results of video and lab experiments in [3].

[1]Videos and code provided in supplementary materials, and available at: `https://anonymousresearcher493.github.io/navigation-prosody-styles.github.io/`



(a) Hesitant motion features are accurately reproduced.

(b) Without the two-level collision constraint, confident motion features are not conserved.

(c) With the two-level collision constraint, confident motion features are accurately reproduced.

Fig. 2. Preservation of hesitant and confident motion features throughout re-planning cycles under sensor noise.

A. *Motion feature reproduction in dynamic environments*

In this subsection, we show several graphs of the velocity commands resulting from deploying our algorithm on a real mobile robot to navigate towards a moving person. We show how the style of motion produced by our approach changes according to the selection of different prosody constraints, and stays consistent even under online re-planning at $10hz$.

Firstly, the algorithm is configured to use the *hesitant* movement prosody constraints (Table I). Fig. 2a shows the commanded velocity profile generated by our algorithm (blue, dashed line). The unfiltered odometry values from the motor encoders are also displayed (orange, solid line), demonstrating that the algorithm generates commands which account for the robot's hardware capabilities. The profile incorporates the saccade motion feature while maintaining the correct average acceleration slope and maximal velocity. Despite the online re-planning to account for the person's motion, the hesitation motion feature is accurately reproduced by the appropriate duration of deceleration and acceleration sequences at time $t = 84.5s$.

Secondly, the algorithm is configured to use the *confident* movement prosody constraints (Table I). In Fig. 2, we show how small amounts of sensor noise or unpredictable person movement impact our ability to accurately produce the desired motion features, and how our two-level collision constraint hierarchy alleviates the issue. At the start of the motion, the acceleration slope is accurately reproduced and the maximum velocity constraint of $0.75m.s^{-1}$ is not violated. When using a single collision constraint (Fig. 2b), a slight deviation between the expected and true position of the obstacle results in an infeasible optimization problem during re-planning. This results in a sudden deceleration at $t = 8.5s$ to ensure safety, which completely changes the motion. In contrast, using our proposed two-level constraint hierarchy (Fig. 2c) enables the *confident* motion features to be preserved by allowing the lower priority collision constraint to be dropped.

Lastly, we demonstrate a reaction to large unplanned movement of a person obstructing the robot's path. Fig. 3 shows the velocity profile generated when using the high kinematics, smooth variant, and hesitation motion sequence

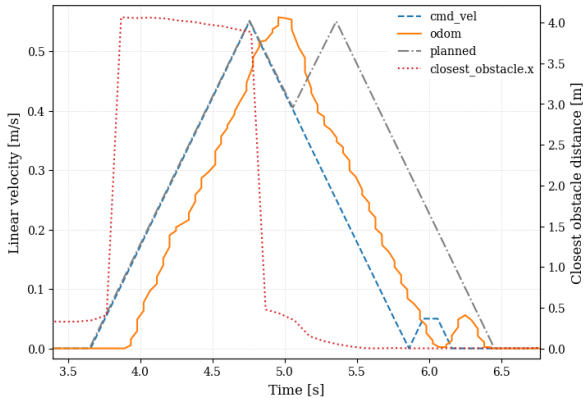


Fig. 3. Partial preservation of motion features in response to large unplanned dynamic obstacle motion.

constraints. Initially, the robot has an obstacle-free path to the goal, and is able to plan a velocity profile that satisfies all prosody constraints, including the hesitation requiring it to slow down and accelerate again (gray, dash-dot line). At time $t = 4.7s$, a person steps in front of the robot, reducing the estimated distance to the closest obstacle (red, dotted line). Without our proposed constraint hierarchy, our algorithm would fail to find a plan satisfying all constraints, and the robot would have to perform an abrupt emergency stop, violating the hesitation, kinematics, and smooth constraints. Instead, our algorithm re-plans a velocity profile that violates only the hesitation constraint (blue, dashed line), enabling the robot's motion to match the desired navigation prosody style much more closely. These results show that our proposed prosody constraints and constraint hierarchy enable accurate reproduction of the navigation movement prosody features proposed in [3] in dynamic environments.

B. *In-the-wild experiment*

In the previous subsection, we demonstrated our algorithm's ability to reproduce the navigation prosody features of [3] in real-time. In this section, we present the first study of these features' effect on HRI in ecological conditions (Fig. 4). The robot was deployed in a university entrance hall over the course of two weeks, where it approached passing

students to hand out fliers. When a person was detected, the robot emitted a greeting sound. If the person stopped, the robot performed an in-place rotation to orient itself towards them. Then, the robot approached the person in a straight line using sets of constraints that prior work [3] associated to *confident* or *hesitant* social attitude perceptions (Table I). While this reproduces the physical set-up of the robot approach used in [3], it differs in that the interaction is spontaneous, and the robot has a defined social role and purpose. An experimenter then conducted a semi-structured interview, initially asking about the robot's performance before explicitly asking about motion and social attitude perceptions. A total of $n = 35$ participants were interviewed (19 for the confident condition, 16 for hesitant). Participants provided informed consent, and all data was stored and processed anonymously.



Fig. 4. In-the-wild flyer handout scenario.

In order to establish whether any difference between the two sets of motion features was perceived, we reviewed participant's responses. When giving their initial impressions of the robot, 63% of participants in the confident condition spontaneously made negative comments about the robot's motion, compared to only 6% in the hesitant condition ($p < 0.01$, two-sided fisher exact test). After the interviewer revealed the purpose of the experiment and explicitly asked participants about the robot's motion, a similar trend emerged with 37% of participants giving negative comments in the confident condition compared to 25% in the hesitant condition ($p < 0.05$, two-sided fisher exact test). Comments referenced the speed or acceleration being too high, or the suddenness of the robot's approach: *"It should slow down more when it approaches the person, it looks like it's going to jump on me. It should decelerate more progressively"*. Another comment, *"It hands out fliers as* ***aggressively*** *as a person would in the street."* spontaneously referenced another social attitude which was found to be correlated with the confident attitude in [3]. These results indicate that the motion generated under the two sets of prosody constraints was perceived differently by participants.

To assess social attitude perceptions, participants were asked to choose between describing the robot as confident or hesitant, which were the expected perceptions for these motion features [3]. The algorithm setting using confident prosody constraints was mostly perceived as confident, with only 1 participant out of 19 perceiving it as hesitant. On the other hand, the hesitant setting was perceived as confident by 12 out of 15 participants ($p < 0.01$, McNemar's test). When explaining why the robot seemed confident, the most common theme, occurring in 51% of responses, was the robot's decision making or attention, stating that the robot seemed to "*decide*" to "*lock-on*" to the person it approached, and "*never changed its mind*". The second most common theme, occurring in 31% of responses, was the robot's path shape, with comments such as "*it went straight*" or "*it came directly towards me*". Interestingly, even participants who perceived the robot as hesitant also mentioned they considered the straight path to indicate confidence.

V. Discussion

The proposed constraints and their hierarchical application in a receding horizon control framework successfully reproduced distinct motion features of social attitude expression [3] in dynamic environments, opening the path to testing the effects of these features outside of video and lab studies. The range of possible experimental scenarios would be improved by extending the approach to fully handle the rotational degree of freedom, however it is currently unknown whether the prosody features proposed in [3] for linear velocity would also be relevant for angular velocity. Other works have shown the importance of robot arm motions in some specific interaction scenarios such as conveying politeness while serving [25], suggesting a similar approach may be relevant to move towards holistic understanding and control of the overall robot motion style.

Differences in positive and negative comments on the robot's approaching motion demonstrate that the two motion styles were perceived differently, however the motion style meant to induce hesitant perceptions only resulted in 5% of participants rating the robot as hesitant. This is in spite of each motion feature being associated with hesitant ratings by more than 70% of participants in prior video and lab-based approach experiments [3]. Unlike the prior work, our experiment involved a robot operating with a clear social role and purpose to hand out fliers. Our experiment also included a pre-approach phase where the robot oriented itself towards a person, which participants seemed to associate to concepts of attention and decisiveness. This shows that the motion features studied in [3] should be combined either with robot orientation features or moderated by the robot's social role in the interaction in order to determine how people will ascribe social attitudes to the robot, calling for further experimentation to understand the link between navigation prosody and perceptions of robots.

One limitation of our HRI experiment is its short interaction duration. Prior work on vocal prosody found that features involved in social attitudes alter people's interactions with robots over longer durations [26], which may also apply to the navigation prosody features.

VI. Conclusion

This work presents an approach to enable newly proposed navigation prosody motion features [3] to be incorporated into mobile robot navigation in dynamic environments in order to study their effect on HRI in ecological experiments that cannot be achieved with hard-coded motions. These features previously associated with social attitude perceptions were formalized as a hierarchy of constraints within a receding horizon control algorithm. We demonstrate that the proposed algorithm ensures accurate reproduction of the prosody features by enforcing consistency of past, current and future planned motion. We use our algorithm to conduct the first experiment testing these features in an ecological experiment, showing that although participants perceived differences between two motion feature sets, these did not fully align with prior lab experiment results reported in [3]. The results suggest that robot orientation dynamics and social role during the interaction should also be considered alongside the navigation prosody features when modeling how people ascribe social attitudes to mobile robots. As future work, the model linking robot motion, robot social role, and perceptions of social attitudes could be refined iteratively in conjunction with the navigation algorithm to improve our understanding of social navigation.

References

[1] C. Mavrogiannis, F. Baldini, A. Wang, D. Zhao, P. Trautman, A. Steinfeld, and J. Oh, "Core challenges of social robot navigation: A survey," *J. Hum.-Robot Interact.*, vol. 12, no. 3, apr 2023.

[2] G. Lombardi, J. Zenzeri, G. Belgiovine, F. Vannucci, F. Rea, A. Sciutti, and G. Di Cesare, "The influence of vitality forms on action perception and motor response," *Scientific Reports*, vol. 11, no. 1, p. 22576, Nov. 2021, publisher: Nature Publishing Group.

[3] P. Scales, V. Aubergé, and O. Aycard, "From vocal prosody to movement prosody, from hri to understanding humans," *Interaction Studies*, vol. 24, no. 1, pp. 131–168, 2023.

[4] A. Moon, C. A. C. Parker, E. A. Croft, and H. F. M. V. der Loos, "Design and impact of hesitation gestures during human-robot resource conflicts," *Journal of Human-Robot Interaction*, vol. 2, 9 2013.

[5] F. Vannucci, G. Lombardi, F. Rea, G. Sandini, G. Di Cesare, and A. Sciutti, "Humanoid Attitudes Influence Humans in Video and Live Interactions," *IEEE Access*, vol. 12, pp. 118 502–118 509, 2024.

[6] K. Mizumaru, S. Satake, T. Kanda, and T. Ono, "Stop doing it! approaching strategy for a robot to admonish pedestrians," in *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2019, pp. 449–457.

[7] A. Francis, C. Pérez-D'Arpino, C. Li, F. Xia, A. Alahi, R. Alami, A. Bera, A. Biswas, J. Biswas, R. Chandra, H.-T. L. Chiang, M. Everett, S. Ha, J. Hart, J. P. How, H. Karnan, T.-W. E. Lee, L. J. Manso, R. Mirsky, S. Pirk, P. T. Singamaneni, P. Stone, A. V. Taylor, P. Trautman, N. Tsoi, M. Vázquez, X. Xiao, P. Xu, N. Yokoyama, A. Toshev, and R. Martín-Martín, "Principles and Guidelines for Evaluating Social Robot Navigation Algorithms," *ACM Transactions on Human-Robot Interaction*, vol. 14, no. 2, pp. 1–65, June 2025, publisher: Association for Computing Machinery (ACM).

[8] L. Raggioli, R. Esposito, A. Rossi, and S. Rossi, "Exploring the Role of Robot's Movements for a Transparent Affective Communication," *IEEE Robotics and Automation Letters*, vol. PP, pp. 1–8, Jan. 2025.

[9] L. Chan, B. J. Zhang, and N. T. Fitter, "Designing and validating expressive cozmo behaviors for accurately conveying emotions," in *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*, 2021, pp. 1037–1044.

[10] S. Song and S. Yamada, "Designing expressive lights and in-situ motions for robots to express emotions," *HAI 2018 - Proceedings of the 6th International Conference on Human-Agent Interaction*, pp. 222–228, dec 2018.

[11] A. Zhou and A. D. Dragan, "Cost Functions for Robot Motion Style," in *IEEE International Conference on Intelligent Robots and Systems*. Institute of Electrical and Electronics Engineers Inc., dec 2018, pp. 3632–3639.

[12] G. Venture and D. Kulić, "Robot Expressive Motions," *ACM Transactions on Human-Robot Interaction*, vol. 8, no. 4, pp. 1–17, nov 2019.

[13] P. Scales, O. Aycard, and V. Aubergé, "Planning socially expressive mobile robot trajectories," *Sensors*, vol. 24, no. 11, 2024.

[14] T. Hielscher, A. Bulling, and K. O. Arras, "Interactive Expressive Motion Generation Using Dynamic Movement Primitives," Oct. 2025, arXiv:2504.06735 [cs].

[15] A. Sripathy, A. Bobu, Z. Li, K. Sreenath, D. S. Brown, and A. D. Dragan, "Teaching robots to span the space of functional expressive motion," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 13 406–13 413.

[16] S. Samavi, J. R. Han, F. Shkurti, and A. P. Schoellig, "SICNav: Safe and Interactive Crowd Navigation Using Model Predictive Control and Bilevel Optimization," *IEEE Transactions on Robotics*, vol. 41, pp. 801–818, 2025.

[17] A. Stratton, K. Hauser, and C. Mavrogiannis, "Characterizing the complexity of social robot navigation scenarios," *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 184–191, 2025.

[18] N. Hirose, D. Shah, A. Sridhar, and S. Levine, "Sacson: Scalable autonomous control for social navigation," *IEEE Robotics and Automation Letters*, vol. 9, no. 1, pp. 49–56, 2024.

[19] Y. F. Chen, M. Everett, M. Liu, and J. P. How, "Socially aware motion planning with deep reinforcement learning," *IEEE International Conference on Intelligent Robots and Systems*, vol. 2017-Septe, pp. 1343–1350, 2017.

[20] H. Khambhaita and R. Alami, "Viewing Robot Navigation in Human Environment as a Cooperative Activity." Springer, Cham, 2020, pp. 285–300.

[21] D. Song, J. Liang, A. Payandeh, A. H. Raj, X. Xiao, and D. Manocha, "VLM-Social-Nav: Socially Aware Robot Navigation Through Scoring Using Vision-Language Models," *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 508–515, Jan. 2025.

[22] N. Campbell and P. Mokhtari, "Voice quality: The 4th prosodic dimension," in *Proc. 15th Int. Congr. Phonetic Sciences*, 08 2003, p. pp. 2417–2420.

[23] M. Schröder, "Experimental study of affect bursts," *Speech Communication*, vol. 40, no. 1-2, pp. 99–116, apr 2003.

[24] Y. Sasa and V. Aubergé, "SASI: perspectives for a socio-affectively intelligent HRI dialog system," in *1st Workshop on "Behavior, Emotion and Representation: Building Blocks of Interaction"*, Bielefeld, Germany, Oct. 2017.

[25] K. Makita, D. Brščić, and T. Kanda, "How Should a Robot do Indirect Handovers Politely?" *International Journal of Social Robotics*, vol. 17, no. 8, pp. 1501–1515, Aug. 2025.

[26] V. Aubergé, Y. Sasa, N. Bonnefond, B. Meillon, T. Robert, J. Rey-Gorrez, A. Schwartz, L. Antunes, G. De Biasi, S. Caffiau, and F. Nebout, "The EEE corpus: socio-affective "glue" cues in elderly-robot interactions in a smart home with the EmOz platform," in *Proceedings of the 5th International Workshop on EMOTION, SOCIAL SIGNALS, SENTIMENT & LINKED OPEN DATA (ES3LOD 2014)*, Reykjavik, Iceland, 2014, pp. 27–34.